

# STATghost

Quick manual

clipboard → local engines → text/plot · v0.4.19 · 2026

*A sniper, not a Swiss army knife.* No editor; clipboard + R / Python / Julia.

## 1 30-second start

1. Unzip so `statghost.exe`, `statghost_rhost.exe`, `lexer/`, and `library/` stay together (rhost only for R.dll; `library/` for Explorer companion).
2. Run `statghost.exe` (toolbar, top-right).
3. Under **Set** → **Session**, pick language **R**, **Python**, or **Julia** and (if needed) the engine path.
4. Click **Idle** → **Armed** (amber).
5. In another app, select code and copy (**Ctrl+C**) — or use the optional Capture hotkey with *no* selection (see below).
6. See results in **Console** / **Plot**.

## 2 Traffic lights

- **Set** — preferences (Appearance includes toolbar icon size 16 px / 24 px / 32 px, default 24).
- **Idle/Armed** — clipboard sniper on/off.
- **Kill** — stop *this app's* current engine child (see below).
- **Panel** — show/hide the output window (**Separados** float bar only; not duplicated on the Unidos tab strip).
- **Clear** — clear console and plot.
- **Unite/Split** — Separados ↔ Unidos (rightmost on the panel tab bar; switch glyph).

**STATghost** icon: menu *Help / About / OS global hotkeys / Quit* (click the icon). Drag the bar by the background or the name. Status shows engine badge and HK: on|off. Chrome glyphs are flat monochrome, tinted for the active theme.

## 3 Engines (R / Python / Julia)

Under **Set** → **Session** (Path | Options):

- **Use** — **R** | **Python** | **Julia**.
- **R**: home (empty = newest on this PC); Use {`Rterm.exe` | `R.dll`}. R.dll talks via sidecar `statghost_rhost.exe` (GUI never loads R.dll).
- **Python**: interpreter path (empty = discover); Options — show Matplotlib window; prefetch imports on session start.
- **Julia**: path (empty = discover / juliaup); Options — show GR window when relevant. Plot panel PNGs use the same high-res size class as R/Python (Plots.jl `savefig` of `current()`).
- **R Options**: show `rgl` window when relevant.

Status badge examples: R.dll · 4.5.0, Python · 3.12, Julia · 1.11, or idle.

## 4 Kill

- Ends only STATghost's current child — Rterm.exe, statghost\_rhost.exe, Python, or Julia — not RStudio / other IDEs.
- Works mid-eval (hard terminate). Toolbar returns to green Idle; click **Arm** for a fresh session.
- Changing language / paths / R Use while running kills the old session first.

## 5 Clipboard and dedup

When **Armed**, eval runs only if clipboard *text* changes. Copying the same selection again does **not** re-run (feature, not a bug). To repeat: change one character, or use a dedicated Capture hotkey (if set).

### Current line without selection

Under **Set** → **Hotkeys** → **Capture**, record an **alternative** global chord (optional). That hotkey asks the foreground app to **Copy**, then STATghost evaluates the clipboard. In many editors (VS Code, Notepad++, RStudio, CudaText, ...), Copy with **no selection** copies the **current line** — so you can step a script line by line without selecting. Behaviour depends on the source app; if Copy does nothing without a selection, select text as usual (or use **Ctrl+C** when Armed).

Operational notices go to the console meta stream; the status bar shows the engine badge and short tips (Console|Plot status font is slightly larger than the system default). After each eval, Console scroll can sit the new chunk at **Middle** (default), Top, or Bottom (Set → Behavior).

## 6 Panel

- **Console** — read-only **ATSynEdit** with Cuda lexers from the `lexer/` folder beside the EXE (R / Python / Julia; packs come from the private companion `statghost-cudatext` at build time). R / Python / Julia use the STATghost Notes markers `#.` / `#..` / `#...` (R also has Programming / Datasets in the slim pack). Mixed text+plot samples use `#. --- GRAPHIC OUTPUT ---` as the two-stage marker (Note1 colour; TF also accepts the classic `# ---` form). **Ctrl+Tab** / **Shift+Ctrl+Tab** switch Console ↔ Plot. The active tab highlight follows the page after eval focus (Auto / Console / Plot). Echo prompts: R `>/+`, Python `>>>/...`, Julia `julia>/julia|` (continuation is classroom-style). Under the Console: small tweaks only inside an echoed **input** chunk (prompt locked; not a free editor), then **Ctrl+Enter** to re-eval (Settings → Shortcuts; Capture stays **Ctrl+C** when Armed). **C+Up/C+Down** (hold letter C, not Ctrl) jump previous/next input cell.
- **Chrome (Separados | Unidos)** — **Unite/Split** button on the tab bar (rightmost). **Separados**: floating traffic bar + panel; lights only on the float (panel shows the mode toggle). **Form único / Unidos** (default): one window — strip on the tab bar (brand | Set | Arm | Kill | Clear); floating bar hidden while the panel is open. Horizontal toolbar magnet: shared left edge under the bar; vertical: tops aligned, rail to the left of the panel.
- **Always on top** — Main traffic bar and Panel share one toggle (default on; chrome / toolbar). Uses Win32 topmost; Explorer has its own Top/Norm control.
- After eval, focus is **Auto** by default: **Plot** if this eval drew graphics, else **Console** (Set → Behavior to force Console, Plot, or return to source).
- **Plot** — high-res PNG (R, Python Matplotlib, Julia Plots.jl); zoom, history ←/→, erase-history

control after the N/M counter, optional PDF (R). Set → Plot panel → Display chooses **Lazarus TImage** (default) or **ATSynEdit Gaps** (same AT family as Console); switch anytime.

- **Size** (Plot panel) — match plot to pane width (default on); PNG DPI slider (soft green → dark blue, default 288) shared by TImage and AT Gaps.
- Bookmarks (if enabled): `Ctrl+Shift+0..9` / `Ctrl+0..9`.

## 7 Window placement

Main, Panel, and Explorer store screen bounds in `%LOCALAPPDATA%/STATghost/window_geometry.json` (separate from prefs). Positions update live on move/resize (short debounce) and restore across sessions, including a float Explorer on a secondary monitor. Stale `(0,0)` / unset `(-1,-1)` captures are ignored.

## 8 Explorer (R)

Object browser for the active R session (VP-EX-1). Requires `library/STATghostcom_R/STATghostcom` beside the EXE (shipped in the zip) or the installed companion under AppData.

- **Open / hide:** traffic-bar **Expl** light, Unidos chrome button, or **Ctrl+E** (configurable under Set → Hotkeys).
- **Dock:** classroom default **Inside** (split the Console tab only; Plot full-width) with **L|R** relative to the editor (`explorer_inside_side / explorer_inside_width`). Or **L | R** beside the main window (dock-left/dock-right), or **Float**. **Top/Norm** stay-on-top apply to External/Float only (Inside never topmost; default Top); **Auto/Fill** height (**Fill** keeps the Explorer status bar inside the work area); window size and Environment pane height are remembered (default eight package rows visible).
- **Environment:** `search()` list with **.GlobalEnv** always first. Click a row to list Objects in that environment. Icon actions: **Detach** selected package:... (`detach(..., unload=TRUE)`), **Load** an installed package (name prompt → library), **Info** (packageDescription). **Hid/All** toggles hidden names (`all.names`).
- **Objects:** exclusive **Tree | Grid** (Name / Type / Class / Dim; Name fills remaining width). Filter by structural type (not S3/S4; icons match Tinn-R order) and optional Perl-compatible name regex (empty box; hint on hover; Enter or leave field stores MRU as `history_explorer_name_filter.json` under AppData STATghost; Up/Down recall). Refresh after eval (when chrome **Auto-refresh** is on, default) or the Refresh button. Objects strip: one-click Help · Example · Plot · Names · Structure · Content · Summary; eraser clears the name regex filter. **Clear** removes visible names from `.GlobalEnv` (confirm). Right-click an object for the Tinn-R-style popup (**Help/Example/Plot/str/Content/...**).
- **Theme:** Explorer follows Settings theme (whole form + Objects popup glyphs).
- **Companion:** R package `STATghostcom` under `library/STATghostcom_R/STATghostcom` — installed on first use into `%LOCALAPPDATA%/STATghost/library`; helpers live in exported `.sg` (no attach). Licence **MPL-2.0** (same as STATghost; LICENSE beside the package sources).

## 9 Settings

- **Appearance** — opacity; **Toolbar** orientation (Separados only — unmarked/disabled under Form único); **Layout** (**Separados | Form único**, default Form único; same as Unite/Split); **Icon size** (**16 px / 24 px / 32 px**, default 24); theme. Chrome uses flat monochrome glyphs tinted for the active theme (traffic bar, Unidos strip, Plot bar, Explorer).

- **Behavior** — after-eval focus (Auto / Console / Plot / source); Console scroll after eval (Middle / Top / Bottom); Console echo — optional echo of full-line # comments in input chunks (default off).
- **Session** — Path | Options per language (see Engines); Ctrl+Tab / Shift+Ctrl+Tab cycle R|Python|Julia pages (does not change Active Use radios).
- **Hotkeys** — OS global: Capture / Arm / Focus / Clear console (**Record** and Clear|Reset). Capture default stays **Ctrl+C** when Armed; optional alternative chord (Copy in the source app → eval; often the current line with no selection — see Clipboard). **Ctrl+L** clears Console text only (plot unchanged); toolbar Clear still wipes console + plot. While Settings is open, OS global hotkeys are suspended so **Record** can capture the same chords (e.g. Ctrl+L). Alt+Space is rejected (Windows system menu).
- **Shortcuts** — Console-local (not OS RegisterHotKey): **Ctrl+Enter** re-evals the input cell at the caret; **C+Up/C+Down** (hold letter C, not Ctrl) jump previous/next input cell (both configurable).
- **Editor** — font, highlighter, view options (Settings Syntax preview mirrors ATSynEdit Console chrome; live Console uses ATSynEdit + lexer/). Under Syntax → Console: chrome **Moldura (frame)** (default) or **Zebrado (zebra)**; optional frame colour, fill, and small corner chamfer. Moldura outlines echoed input only. Ctrl+Tab / Shift+Ctrl+Tab cycle Font|Syntax.
- **Plot panel** — Display (TImage | AT Gaps), history ←/→ + erase-history, PDF, Size (match pane + DPI).

## 10 Hotkeys

- **Ctrl+Alt+R** — Arm / Idle (default; Record under Set → Hotkeys).
- **Ctrl+C** — Capture via system clipboard (when Armed).
- Optional Capture alternative (Record under Set → Hotkeys) — Copy in the foreground app, then eval (current line without selection in many editors). Active **only while Armed** (Idle does not register or fire send/eval chords).
- **Ctrl+Alt+F** — Focus mode toggle (default).
- **Ctrl+L** — Clear Console text only (default; R console habit). The Console ATSynEdit keymap does **not** keep the stock Ctrl+L “extend selection by line” binding (would steal Clear).
- **Ctrl+E** — show / hide Explorer (default).
- Brand menu **OS global hotkeys** — when off, Arm / alt-capture / focus / clear / Explorer chords are unregistered so other apps can use them; status shows HK: ON/HK: OFF. Ctrl+C sniper while Armed still works. Opening Settings also suspends those OS chords until the dialog closes (so Record can hear them).

## 11 Shortcuts (Console)

Local to the Console panel (not OS RegisterHotKey). Set → Shortcuts.

- **Ctrl+Enter** — re-eval the input cell at the caret.
- **C+Up / C+Down** — previous / next input cell (hold letter C, not Ctrl — keeps Ctrl+arrows for scroll).

## 12 Config

Prefs JSON: %LOCALAPPDATA%\STATghost/statghost.json.

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## 13 Automated checks (lab)

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- **Smoke:** `statghost.exe --smoke` → `statg5_report.txt` (RESULT=BOK).
- **Arm on start:** `statghost.exe --arm` (Armed + session up — handy after a rebuild).
- **SR / SP / SJ / ST:** whole-file samples — `--sr` (R `rterm+rdll`), `--sp` (Python), `--sj` (Julia), `--st` / `--tf-samples` (all) → `statg_sr|sp|sj|st_report.txt`. Or: `powershell -File src/tf_samples.ps1 [-Mode ST|SR|SP|SJ]`.
- **Dev smokes (SRD|SPD|SJD|STD):** random ~20% of scripts per lane (`--srd...--std`). Prefer these while iterating.
- **BENCH:** same OLS lesson in all four environments (Rterm.exe | R.dll | Python | Julia) — `--bench` → `w_log/benchmark.txt`; `powershell -File benchmark/run_bench.ps1`.

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## 14 Troubleshooting

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- No output — **Armed?** New clipboard text? Correct language under Session? Badge not idle?
- No plot — this chunk must draw (`plot`, `Matplotlib`, `Plots.jl`, ...).
- R.dll won't start — is `statghost_rhost.exe` next to the app? Is R.dll under R home? Or switch Use to `Rterm.exe`.
- Python / Julia not found — set path under Session → Path, or install and leave empty for discover.
- Stuck eval — press **Kill**, then Arm again.
- Hotkeys clash with another app — turn off **OS global hotkeys** (sniper `Ctrl+C` when Armed remains).
- Taskbar icon looks outdated after an update — unpin / repin `statghost.exe` (Windows caches MAINICON).